Guitar Explorer — Black Dog Song Study

Functional Requirements Document (FRD) · Version 0.1 · Review Draft

Document control	Value
Project	Guitar Explorer — Song-Led Musical Discovery
Pilot	SS-001: *Black Dog* — Led Zeppelin, original studio recording
Status	**Draft for product-owner and musical-source review; NOT approved or implementation-authorizing**
Prepared	11 November 2026
Planning authority	GitHub Issue #83; Trello Song Study milestone
Repository reference	PR #84, exact head 8a779419808fb94317ca83ad454fd548887094d8, **open/unmerged at drafting**
Research inputs	Argon *Phase 1 Product Requirements Research* (51-page PDF) and *Targeted Correction Pass* (Markdown, 11 Nov 2026)
Intended readers	Product owner, musical researcher, design researcher, Codex/engineering after separate authorization

> **Document rule:** This FRD records the desired learning outcomes and testable product behavior. It does not approve new code, a new data schema, any change to the deterministic Music Engine, or merging PR #84. Numerical song timings and detailed harmonic interpretations from Argon's research remain candidates until independently checked against the selected performance.

---

1. Executive requirements decision

Guitar Explorer will pursue **song-led musical discovery**. A learner begins with a song they love, learns how its parts fit together, investigates meaningful musical relationships in the music, plays them on a physical guitar, and tests whether they can use those relationships elsewhere.

The primary failure to avoid is recreating the recently retired collection of independent drills, merely replacing each exercise title with a song title. The *Black Dog* pilot must feel like investigating one continuous piece of music: the same motifs, pulse, changing accompaniment and guitar decisions recur across sections.

Two outcomes are distinct and both matter:

1. **Song fluency:** recognize, navigate, learn and physically play the relevant parts of the arrangement.
2. **Musical independence:** hear, explain, locate, vary and transfer at least one musical idea from the song to another musical context.

The selected product approach is a **working hypothesis** of song-first learning with optional extracted musical principles. Principle Cards, personal libraries, exercise archetypes and automated cross-song discovery are not yet approved implementation requirements.

Pilot scope: the whole studio recording is in scope for requirements, source mapping and the journey. The **first implementation candidate** is one short main-riff chromatic-approach investigation; the E-Turnaround and outro solo remain mapped follow-ons. A larger 0:06–0:52 three-investigation suite was proposed by Argon but is **not accepted as the first engineering scope**.

2. Goals, users and non-goals

2.1 Primary user and job

A guitarist can follow a practical tutorial and learn fret positions, but does not want to finish another song with only finger memory. During study they may have their guitar in their lap, frequently pause the recording, and need one musical question at a time.

Job to be done: “Help me understand the specific things that make this song work, see and hear those relationships on my guitar, and learn to use them myself.”

2.2 Objectives

- Make the **whole arrangement intelligible**, including recurring sections and what changes between occurrences.
- Connect real auditory material with deterministic theoretical facts **without treating interpretation as mathematical fact**.
- Support physical fretboard decisions instead of only screen responses.
- Help learners compare authentic/source-backed material with **explicitly labeled original educational variations**.
- End investigations with transfer, while allowing the learner to revisit or skip explanations.
- Keep the UI focused and minimal; controls appear only where the musical question needs them.
- Allow reuse across a second contrasting song **without committing to universal exercise templates** in advance.

2.3 Explicit non-goals for this FRD / first slice

Not a TAB piracy or transcription service; not an automatic guitar grading system; not a generative style-pack platform; not a DAW or multitrack editor; not a universal curriculum, mastery model, router or generic Exercise DSL; not a full-scale song importer; not a new Music Engine rewrite; not automatically rebuilding retired experiences; not authorizing a framework of seven implementation templates.

3. Sources, provenance and evidence status

3.1 Source register

Source ID	Artifact	Purpose	Evidentiary limitation
SRC-01	Led Zeppelin — *Black Dog*, official studio audio	Performance, arrangement, audible phrasing and occurrence timing	External player times may vary by version; detailed transcription still needs human listening validation
SRC-02	Marty Music — *Black Dog* guitar lesson, 2016	Learner's confirmed practical playing guide	Tutorial fingering is not proof of Page's exact recorded fingering; it does not cover all parts
SRC-03	StephLeGratteux — outro solo video	Third-party solo notation/playing reference	Arranged ending and performer/transcriber choices may differ from studio fade-out
SRC-04	User-supplied **four-page** *Black Dog guitar solo (outro solo)* PDF	Notation, TAB, technique marks, positions; available to this drafting workspace	Not note-by-note verified against original; not accessible to Argon in its correction pass
SRC-05	John Paul Jones retrospective/interview material	Artist testimony on rhythm and arrangements	Reported intent; must not be conflated with independently measured score timing
SRC-06	Aaron Krerowicz — rhythmic displacement analysis	Nine-eighth-note pattern against 4/4	Short analytical text, not full transcribed arrangement
SRC-07	The Music Professor — synchronized rhythm analysis	Beat/pickup and phrase-grouping hypotheses	Requires checking counting convention and editorial notation against the recording
SRC-08	Marin Music Center — alternate *Black Dog* tutorial	Alternative riff fingering; bridge section	Secondary tutorial; not proof of studio left-hand fingering
SRC-09	GuitarForce — Verse 3 harmony lesson	Candidate overdubbed harmony material	Third-party teaching analysis to corroborate
UX-01	Guitar Jams Together video reference and user screenshot	Clear, single musical mission; backing / current musical context	Visual design input, not pedagogical outcome evidence
UX-02	StrumForge and user screenshot	Relevant playback, scale, arpeggio and exercise affordances	Feature inventory rather than a validated Song Study journey

Source ID	Artifact	Purpose	Evidentiary limitation
RES-01	Argon 51-page Phase 1 report	Broad candidate facts, UX proposals, FR-01–18 and NFR-01–09	Superseded where contradicted by correction pass and verified engine source
RES-02	Argon correction-pass Markdown	Corrected section map, duration units, theory labels and citations	Some “verified” labels are Argon's, not independent review by the FRD author
REPO-01	PR #84 head: src/engine/core/phrase.ts, src/engine/guitar/fretboard.ts, src/research/types.ts, src/research/contract.ts	Verified technical boundary at exact proposed reset commit	Branch unmerged; not a working deployed Song Study product

Source hierarchy: recording evidence and observable source features → reliable source claims → engine-computed facts under explicit assumptions → musical interpretation → product proposal. In all stages, source confidence and the distinction between *observed*, *reported*, *estimated*, *contested* and *unknown* must be retained.

3.2 Musical claims awaiting independent validation

- **Eleven macro-sections / 31 atomic events:** Argon's corrected editorial decomposition. It is coherent as a draft but not a natural or hard-coded music fact; event count depends on granularity.
- **Timestamps** below: approximate coordinates as supplied by Argon, not final syncpoints.
- **Bridge harmony:** the report's “F#m6” name is **not accepted**. The cited pitch collection does not establish that chord; use an A-centered / F#m-related color only as a source-labeled analytical hypothesis until checked.
- **Main riff:** use a reported A-minor-pentatonic foundation with a G# chromatic approach and expressive C inflection, not a blanket “A Dorian” label.
- **E-Turnaround:** keep actual sounding pitch collection distinct from a functional dominant-chord claim; the reported G-natural is not the major third of E.
- **Chord quality:** a D5 power chord contains D and A, not F#. F# may be a melodic harmonic color or a D-major-third target **if** D-major harmony is separately supported.
- **Exact recorded fret choice:** not inferable from the studio audio alone. Marty and other tutorials are alternative physical realizations.
- **Tempo / meter / beat:** count in explicit quarter-note beats or eighth-note subdivisions; no automatic equivalence between 9 eighth notes and 9 sixteenth notes.

4. Whole-song section map (provisional)

4.1 Macro structure

Section	Estimated studio position	Proposed musical focus	Primary learner discovery	Source basis
SEC-01 Intro	0:00–0:06	Guitar/tape texture before pulse	Can texture prepare a musical entrance?	SRC-01, RES-02
SEC-02 Verse 1 / full call-response	0:06–0:52	Vocal calls, main riff, chromatic approach, E-Turnaround	Pickup vs. downbeat; chromatic gravity; riff displacement	SRC-01/02/05/06/07
SEC-03 Chorus 1	0:52–1:05	Contrasting bend/power-chord gestures	Major/minor blues color without imposing a single scale	SRC-01/02
SEC-04 Verse 2 / shortened	1:05–1:27	Two vocal calls rather than full verse return	Recognize an omitted event and anticipate the next section	SRC-01/08
SEC-05 Bridge 1	1:28–1:39	Contrasting guitar/vocal figure and ensemble entry	How an arrangement changes by removing instruments	SRC-01/08; correction needed for chord name
SEC-06 Chorus 2 / “Hey Baby” groove	1:39–2:03	Repeated chordal groove / turnarounds	Find chord targets in a moving backing context	SRC-01/02
SEC-07 Verse 3 / harmonized	2:04–2:49	Returning riff plus upper guitar harmony	What changes when a second line enters?	SRC-01/09
SEC-08 Chorus 3	2:49–3:02	Return of Chorus 1 material	Hear repetition and micro-variation rather than relearning	SRC-01
SEC-09 Verse 4 / shortened	3:02–3:23	Shortened two-call verse	Hear large-scale form and expectation	SRC-01
SEC-10 Bridge 2 / solo launch	3:23–3:35	Return of bridge; change into lead	How a transition sets up solo phrasing	SRC-01/03/04/08
SEC-11 Outro solo / fade	3:35–4:55	Blues/major-minor coloration, position and motif development	How solo ideas travel through the neck and relate back to the song	SRC-01/03/04

Editorial rule: display recurring material as related occurrences, not independent song “exercises.” The full song is always recoverable from the selected passage.

4.2 Atomic occurrence map

These 31 events are **candidate editorial anchors** from RES-02. Timing, alignment and special-case transitions require listening approval before publication. Sections and events should use stable IDs, but schema validation must **not** enforce this count for other songs.

Event	Parent	Approx. range	Label
EVT-01	SEC-01	0:00–0:06	Intro texture
EVT-02	SEC-02	0:06–0:12	Vocal call 1a
EVT-03	SEC-02	0:12–0:18	Main riff response 1a
EVT-04	SEC-02	0:18–0:24	Vocal call 1b
EVT-05	SEC-02	0:24–0:29	Main riff response 1b
EVT-06	SEC-02	0:29–0:35	Vocal call 1c
EVT-07	SEC-02	0:35–0:41	Extended main riff 1c
EVT-08	SEC-02	0:41–0:46	E-Turnaround 1
EVT-09	SEC-02	0:46–0:52	Main riff return 1d
EVT-10	SEC-03	0:52–1:05	Chorus 1
EVT-11	SEC-04	1:05–1:10	Vocal call 2a
EVT-12	SEC-04	1:10–1:15	Riff 2a
EVT-13	SEC-04	1:15–1:21	Vocal call 2b
EVT-14	SEC-04	1:21–1:27	Riff 2b
EVT-15	SEC-05	1:28–1:39	Bridge 1
EVT-16	SEC-06	1:39–2:03	Chordal groove / chorus 2
EVT-17	SEC-07	2:04–2:10	Vocal call 3a
EVT-18	SEC-07	2:10–2:15	Harmonized riff 3a
EVT-19	SEC-07	2:15–2:21	Vocal call 3b

Event	Parent	Approx. range	Label
EVT-20	SEC-07	2:21–2:26	Harmonized riff 3b
EVT-21	SEC-07	2:26–2:32	Vocal call 3c
EVT-22	SEC-07	2:32–2:38	Extended harmonized riff 3c
EVT-23	SEC-07	2:38–2:43	E-Turnaround 2
EVT-24	SEC-07	2:43–2:49	Harmonized riff return
EVT-25	SEC-08	2:49–3:02	Chorus 3
EVT-26	SEC-09	3:02–3:07	Vocal call 4a
EVT-27	SEC-09	3:07–3:12	Riff 4a
EVT-28	SEC-09	3:12–3:18	Vocal call 4b
EVT-29	SEC-09	3:18–3:23	Riff 4b
EVT-30	SEC-10	3:23–3:35	Bridge 2 / solo launch
EVT-31	SEC-11	3:35–4:55	Outro solo to fade

5. Section-by-section learner journeys

Each chapter below defines a **distinct musical investigation**, not a required uniform step sequence. A learner can always return to the recording, hear the material and play without completing a theory explanation.

Section	Hear / ask	Locate / play	Experiment and transfer	Research guardrail
SEC-01 Intro	What is texture rather than a stable pitched riff?	Try a safely controlled pick scrape or muted percussive gesture	Create an original 2-second entrance to a different riff	Do not claim precise effects chain or pitch contour without evidence
SEC-02 Verse 1	Where does the pickup lead; why does G–G#–A feel directional; how does the E cell sit against drums?	Play the short chromatic climb in Marty's position; locate target A elsewhere	Replace/omit G# in an **original** comparison, then approach another root	Validate pickup/bar alignment; separate beat counting from rubato vocal gaps

Section	Hear / ask	Locate / play	Experiment and transfer	Research guardrail
SEC-03 Chorus 1	What changes between bright bend and darker power-chord color?	Hear and play a short representative bend/response	Compare melody against power-chord vs. separately verified fuller harmony	Bend trajectory and exact major/minor quality require source validation
SEC-04 Verse 2	Why does the third vocal/riff cycle not return?	Play the shortened verse without full-song TAB scrolling	Predict a shortened phrase in an original call-response	Detect formal difference, not abstract ear-quiz scoring
SEC-05 Bridge 1	Why does the energy change when instruments drop away?	Learn a source-labeled guitar phrase and its resolution	Change the number of layers in an original arrangement	Do not label F#m6 as a fact; do not confuse with Chorus 2 groove
SEC-06 Chorus 2	What harmony anchors the repeated groove and turnaround?	Play chord skeleton; find root/fifth and a deliberate melodic target	Create one short target-note response over an original rock progression	D5 vs. D-major harmony must be stated precisely
SEC-07 Verse 3	Which part is new compared with Verse 1?	Learn a short counterline; visualize vertical intervals	Harmonize a short original motif in thirds/fourths	Do not pretend monophonic Phrase v1 already represents polyphony
SEC-08 Chorus 3	What sounds identical and what changes from SEC-03?	Replay familiar passage with eyes off TAB	Change articulation of one note in an original response	Verify exact difference rather than inventing novelty
SEC-09 Verse 4	Can you anticipate the upcoming bridge after another shortened verse?	Play form landmarks from memory	Make a small, original contrasting form with two-call/three-call variants	No mastery claim from correct screen navigation
SEC-10 Bridge 2	What signals solo entry?	Find the pickup/lead transition on the neck	Create a transition from an original rhythm riff into a solo	Solo PDF/video may diverge at boundaries and ending
SEC-11 Outro solo	How does register, motif and target note evolve through the solo?	Trace selected phrases across neck regions; study bends as physical actions	Improvise a short five-zone—or smaller—register story and quote one self-made riff cell	Solo transcription and studio track are separate versions; do not assume exact source fingering

6. Session experience and information hierarchy

6.1 Entry and overview

Song Overview answers: “Which recording am I studying? What are the sections? Where am I? What can I discover?” It should display the song identity, optional linked tutorial/source, the ordered section map, and brief section-specific musical questions. It must not show an entire fretboard, several chord diagrams and audio animations before the learner chooses a question.

Section Workbench answers: “What am I hearing and doing now, and why?” It shows one current musical mission, the relevant contextual controls, one primary work surface, progressive musical explanation where useful, and a way to return to the song overview.

6.2 Proposed visibility contract

UI element	Visibility	Rule
Song/section breadcrumb + musical question	Always while in section	One concrete question, no stacked exercise taxonomy
Reference transport / open-source link	When auditory reference is available	Always clearly distinguish external source and synthetic original example
Current harmony/beat	Only when relevant	One canonical presentation; no duplicated “Now/Next” chord panels
Focused fretboard	For pitch, fingering, technique, or transfer tasks	Relevant fret region first; expand for position/solo questions
Rhythm lane/grid	For timing, pickup, displacement	Hidden for purely harmonic or positional tasks
Focused fragment notation/TAB	When useful and lawfully available	Optional; short selection; avoid full scrolling score as workbench default
“Why it works”	On demand / progressive reveal	Don't spoil the musical question before the learner gets to hear/play
“What changes if…?”	In appropriate experiments	One controlled musical variable at a time
Physical-transfer prompt	At natural end of discovery	Permits an honest self-report, not fictitious automatic checking

Accessibility and hands-busy usability: readable from practical guitar-playing distance, appropriate touch targets, keyboard accessibility, no color-only meaning and no hover-only critical controls. Do not encode hard pixel/layout values as music requirements; validate mobile portrait, mobile landscape and desktop with real prototypes.

7. Functional requirements register

Status legend: Pilot must = needed to validate the first focused learner experience; Whole-song must = required before claiming a complete whole-song study; Next = a later section-specific capability; Later = non-blocking future option. A requirement may exist in the FRD without being in the first engineering PR.

ID	Functional requirement	Priority	Observable acceptance criterion	Dependency / current status	Trace
SS-FR-01	Identify work, arrangement, realization, source and version	Pilot must	The opening passage shows *Black Dog*, studio/source identity and the Marty tutorial as separate references	New product wrapper; existing research contract supplies partial provenance vocabulary	SRC-01/02; REPO-01
SS-FR-02	Show coherent recording-order macro-section map	Whole-song must	User can select each of the 11 provisional macro-sections and return without losing location; verified boundaries replace estimates	New UI and curated mapping; no fixed 11-count schema assumption	RES-02 map
SS-FR-03	Identify repeat occurrences and shared musical materials	Whole-song must	Verse 1/3 and 2/4 can be navigated as linked occurrences with differences labeled	New mapping; editorial review	SEC-02/04/07/09
SS-FR-04	Open focused section and retain return context	Pilot must	Selecting main riff opens a focused mission and returns to its parent recording section	New UI; no old Practice Runtime	BD-INV-01
SS-FR-05	Store source-backed fragment observations separately from interpretation	Pilot must	Every shown sourced note/fingering has source and provenance; contested studio positions remain alternatives	Reuse/extend research layer only as demonstrated; no core engine rewrite	REPO-01; RES-02
SS-FR-06	Ask a musically grounded question from the chosen passage	Pilot must	Mission refers to what the learner can hear and play in the specific passage, not random scale parameters	New bounded content	BD-INV-01
SS-FR-07	Provide source access or lawful reference route	Pilot must	Relevant official recording/tutorial can be opened or cited; unavailable media is disclosed rather than silently substituted	External links first; integration unspecified	SRC-01/02
SS-FR-08	Present focused fretboard note locations and roles	Pilot must	Learner sees short E–G–G#–A phrase in a requested physical region and can distinguish A root, G flat-7 and G# approach	Engine pitch/interval/fretboard queries exist; new renderer required	REPO-01; BD-INV-01
SS-FR-09	Generate musical facts using explicit assumptions	Pilot must	Note/interval facts are derived from the engine and display their local tonal assumption; C# / G# claims are never silently mislabeled	Engine exists; contextual interpretation remains editorial	REPO-01
SS-FR-10	Permit active listening or prediction before explanation	Pilot must	Learner can hear the reference or educational comparison before viewing its analysis; they can also reveal immediately	New UI; no compulsory lesson state machine	BD-INV-01; RES-01 Section F
SS-FR-11	Provide a real-guitar action	Pilot must	Learner is asked to play a precise short phrase, with instructions that work even without microphone permission	New UI/copy	BD-INV-01
SS-FR-12	Compare at least one controlled original educational variation	Pilot must	Learner compares with/without a chromatic approach using a clearly labeled original example and can identify what changed	Requires appropriate small sound/reference solution; engine provides underlying notes, not browser audio	BD-INV-01; licensing review

ID	Functional requirement	Priority	Observable acceptance criterion	Dependency / current status	Trace
SS-FR-13	Transfer the idea outside the source excerpt	Pilot must	Learner gets a different target-root/key or neck region and is invited to play an original approach without fixed TAB; prompt remains valid when hints are hidden	Engine transposition/location queries; new UI	BD-INV-01
SS-FR-14	Separate performance evidence types	Pilot must	App distinguishes user screen response, optional self-reported physical playing and truly observed audio (not present)	New product behavior, no grading	RES-01 Section F
SS-FR-15	Workbench uses conditional views and no redundant chord displays	Pilot must	Chroma passage does not show timing grid/solo map; no duplicate chord diagrams; mobile layout usable	New UI	UX-01/02
SS-FR-16	Support a verified pickup and beat-position explanation	Next	When tested on the recording, the selected riff's pickup and downbeat are displayed against declared eighth-note units	Exact time exists; new alignment model/UI; no negative Phrase event required by default	BD-INV-02; RES-02 rhythm
SS-FR-17	Show phrase displacement against underlying meter	Next	A 9-eighth cell is shown relative to 4/4, with 4.5 quarter-note beats per 9-eighth cycle, and source-checked phase placement	Exact time basis exists; new presentation & authoring	BD-INV-04
SS-FR-18	Explain local harmony and chord roles across section changes	Next	Current note role is computed against an explicit, verified chord; D5 is never mislabeled as having F# chord tone	Engine chord-role support; timed-harmony source/UI missing	BD-INV-06/12
SS-FR-19	Compare alternative physical realizations	Next	Tutorial 7→9 G-string slide and cross-string E4 possibility show same sounding note, with distinct source statuses	Existing geometry; new compare UI	BD-INV-03
SS-FR-20	Handle source-labeled expressive technique	Next	Bend-origin, target and expression are distinguishable, without claiming that Phrase v1 supports continuous bends	New technique representation outside immutable core unless proven necessary	BD-INV-09/solo
SS-FR-21	Compare two musical voices without conflating them	Next	Verse 3 can show primary riff and proposed upper harmony in alignment; intervals computed only where notes are known	New event-alignment layer; Phrase v1 monophonic	BD-INV-07
SS-FR-22	Show section-specific solo phrase/region roadmap	Next	A reviewed passage can highlight target notes across configured frets 0–20; original and transcriber ending distinguished	Configurable engine board; new UI and verified solo excerpts	BD-INV-10/11; SRC-03/04
SS-FR-23	Save a simple transferable musical discovery	Later	Learner may save a plain note linking musical principle to source section; no graph/inference claims	New persistence feature, no immediate need	Model C
SS-FR-24	Import user-supplied symbolic fragments	Later	Validation preserves source, pitch, time, physical location and uncertainty where provided	New import and rights requirements	RES-01 H

ID	Functional requirement	Priority	Observable acceptance criterion	Dependency / current status	Trace
SS-FR-25	Add optional sound-recording or microphone mirror	Later	Consent-based local capture or pitch indication is clearly non-grading and survives permission denial	New audio work; no claim of current support	RES-01 J

Clarification on acceptance: For the first slice, *physical learner value* is assessed through a real guitar session and the owner's feedback, not only programmatic checks. SS-FR-12 does not authorize a three-stem sampler, licensed studio-audio hosting or a full audio architecture.

8. Nonfunctional requirements and safeguards

ID	Requirement	Proposed verification
SS-NFR-01	Deterministic musical correctness and clear assumptions	Existing 552-test local suite remains a baseline as reported for PR #84; engine query smoke tests; independent examples reviewed against engine output
SS-NFR-02	Provenance, uncertainty and competing interpretations	No unqualified “Page played this fret” from audio alone; provenance review and source mismatch examples
SS-NFR-03	Copyright and external-platform compliance	Licensed/permitted materials review before bundling; no ripping recordings, score scraping or assumed safe “four-bar exemption”
SS-NFR-04	Responsive, accessible hands-occupied guitar use	Desktop, mobile portrait/landscape, large controls, keyboard access, labels not color alone
SS-NFR-05	No misleading grading	Manual self-reports visibly separate from app-observed responses and optional future microphone data
SS-NFR-06	Honest media availability	Missing/blocked external reference reports unavailability; synthetic examples labeled original
SS-NFR-07	Engine independence and modularity	No mixing of learner states/audio objects into src/engine/core or query contracts without a separately reviewed proof of missing facts
SS-NFR-08	Runtime performance	Only measure rendering/playback latency if those modules are built; audio sync tested against shared musical clock before claiming precision
SS-NFR-09	Maintainable bounded song authoring	A second contrasting-song pressure test uses the same conceptual work/section/question model without needing a new framework

Evidence caution: the reported 552 passing tests / 17 files is Codex's local verification of PR #84, not an independently rerun result in this drafting task. GitHub Actions/preview publication were blocked separately.

9. Music Engine and existing-research boundary

Existing at reviewed PR #84 head	Not verified as existing / new requirement	Action decision
Spelled pitches, registered notes, intervals, chord/scale/harmony primitives and query APIs	Automatic interpretation of a commercial song	Preserve engine; manually/source-author observations
Standard tuning and validated configurable string/fret coordinate calculations	Browser fretboard visualization and movable region overlays	New UI in later authorized slice; default fretboard is 15 frets, maxFret configurable
Exact rational time (MusicalTime) and immutable **monophonic** Phrase with pitch/onset/duration	Bend metadata, simultaneous polyphony, bar/recording sync, continuous gesture envelopes	Keep technique/timing metadata in an external material layer first; do not assume core modification
Fragment extraction, interval/contour/rhythm analysis, exact transposition	A study of every solo bend and fingering from PDF alone	Incremental source-backed content review
src/research/types.ts + src/research/contract.ts provenance vocabulary	Generic SongStudyPack loader and validated full-song schema	Audit compatibility first; do not create duplicative schema automatically
Engine build and minimal static welcome page on PR #84	Song Overview, Section Workbench, audio transport, local recording	New learner-facing product work after owner approval

Technical notes from source inspection: src/engine/core/phrase.ts rejects events with negative local onset and does not admit arbitrary bend fields; a song-facing coordinate system can instead place the pickup within a nonnegative fragment and separately mark a downbeat offset. src/engine/guitar/fretboard.ts defaults to maxFret: 15, but permits validated higher values, so a 20th-fret solo zone is not in itself justification for expanding engine theory.

10. Initial pilot — approved for requirements discussion only

10.1 Candidate vertical slice: “The Chromatic Launchpad”

Parent section: SEC-02 Verse 1, approx. 0:12 opening riff response (exact source sync to validate). **Targeted fragment concept:** E–G–G#–A leading to root A, described in the Argon research as 5 → ♭7 → 7 → 1 relative to A.

Learner outcome: Identify the target A, hear the effect of the added G# chromatic approach, play the compact idea in a comfortable position, then create a similar approach toward a different root in another position or key.

Experience contract (not an implementation schema):

1. **Hear the song in context.** Open official-source reference with time locator; do not embed an unauthorized audio extract.
2. **Play it.** Use a source-labeled practical fingering if available; no software gate to touching the guitar.
3. **Ask why.** The UI highlights root, flat seventh and major seventh under the explicit A reference, with scale-pentatonic membership distinguished from nonmember chromatic G#.
4. **Try a controlled variation.** Audition or compare a short **original educational** equivalent lacking G#; avoid describing it as an alternative studio take.
5. **Transfer.** Prompt a corresponding leading approach into another chosen root/fret region; offer help, then remove fixed fret guidance.
6. **Reflect.** Self-report what changed in hearing, reasoning and physical playing; don't claim machine-certified ability.

Pilot acceptance tests

Test	Given	When	Then
PAC-01 Source clarity	Studio, Marty and optional other tutorial references	Learner opens the passage	UI identifies separate sources; no false claim that tutorial fingering is the studio fingering
PAC-02 Correct pitch roles	Reported E/G/G#/A and explicit A root	User reveals analysis	Engine-derived 5, ♭7, 7, 1 relationships appear without falsely classifying G# in A minor pentatonic
PAC-03 Relevant fretboard	Short riff in selected region	Learner expands or hides labels	Display shows playable locations and avoids unrelated full-neck clutter
PAC-04 Active comparison	Original educational phrase and labeled altered phrase	Learner toggles A/B	Only the named musical variable changes; audio/reference source is explicit; no deceptive studio attribution
PAC-05 Physical transfer	A different target root or region	Learner attempts new approach	Valid target is provided; learner can attempt without TAB; self-report does not become a graded “pass”

Test	Given	When	Then
PAC-06 Section continuity	Learner in focused question	User returns to overview	The song/section/position remain clear; no unrelated exercise catalogue shown
PAC-07 Fail safely	Reference media unavailable / browser sound blocked	Learner opens investigation	UI provides an honest fallback, does not falsely imply playback
PAC-08 Musical validation	Learner uses guitar with the pilot	Learner completes and discusses experience	Product owner can state a specific new musical insight and reuse attempt; negative feedback recorded and may block merge

10.2 Why the E-Turnaround is not in the first implementation PR

Argon recommended combining three investigations: chromatic riff, beat-one pickup, and 9-over-4 turnaround. Those are valuable **subsequent** learning tasks. However, they require source alignment, different rhythmic representations, potentially audio scheduling and an added stage that could make the first slice too large to learn from. After the riff pilot, choose the **beat-one pickup** or **E-Turnaround phase shift** next on evidence, not on template count.

Corrected rhythm research checkpoint (not yet validated sync)

The reported E-Turnaround cell has note duration groups of 1 + 1 + 1 + 1 + 1 + 2 + 2 = 9 eighth notes = 4.5 quarter-note beats when counting near 148 BPM. A 9+9+9+5 group is 32 eighth notes = 16 quarter-note beats = four 4/4 bars; the 3-eighth-note pickup into the next riff and the choice of reference downbeat must be modeled separately. An equivalent half-time count using sixteenths must be explicitly converted rather than mixed silently. The claimed recording time window needs independent beat/sync checking before any acceptance test asserts its precise event offsets.

11. Traceability: song material → learning question → requirement

Research investigation	Musical question	Proposed requirements	Phase
BD-INV-01 Chromatic Launchpad	What does the chromatic approach do?	SS-FR-04 to 14	Pilot candidate
BD-INV-02 Beat One	Which note lands on the downbeat?	SS-FR-16, 07, 10, 11	Next
BD-INV-03 Alternate fret positions	Why play equal pitch on different strings?	SS-FR-19, 08	Next
BD-INV-04 E-Turnaround	How does 9-eighth grouping move over 4/4?	SS-FR-17, 07	Next

Research investigation	Musical question	Proposed requirements	Phase
BD-INV-05 String-set translation	Why can the pattern shift across strings?	SS-FR-19, 08	Next
BD-INV-06 Major/minor color	What do bends/thirds suggest over power chords?	SS-FR-18, 20	Next
BD-INV-07 Added guitar harmony	How does another voice change a riff?	SS-FR-21	Next
BD-INV-08 Bridge/ensemble layering	Why does removing/restoring accompaniment change character?	SS-FR-03, 18, 21	Next
BD-INV-09 Expressive micro-bend	What is the difference between fretted and bent pitch?	SS-FR-20	Next
BD-INV-10 Solo zone roadmap	How does register movement tell a story?	SS-FR-22	Next
BD-INV-11 Reused riff cell	Does an earlier idea return in the solo?	SS-FR-03, 22, 23	Next; require audio confirmation
BD-INV-12 Turnaround targeting	How does solo line follow moving harmony?	SS-FR-18, 22	Next; preserve chord-quality precision

12. Risks, open questions and approval gates

ID	Open issue	Impact / resolution owner
OPEN-01	PR #84 is still open/unmerged; hosted Actions startup_failure / preview 404 previously reported	Owner must explicitly review exact head and gate; FRD cannot silently override merge policy
OPEN-02	Exact 11-section / 31-event timing map needs human studio listening and verified source sync	Musical source audit, before “verified” labels

ID	Open issue	Impact / resolution owner
OPEN-03	Main riff and E-Turnaround note durations/beat alignment need audio-to-score check	Before implementing precise pulse/loop view
OPEN-04	Bridge “F#m6” label and several modal/major-minor claims remain contested	Musical editor; keep claims conditional
OPEN-05	Studio fretboard fingerings cannot be deduced from audio	Keep tutorial/source variants; no fabricated provenance
OPEN-06	Solo PDF available here, absent from Argon's pass; video may include altered ending	Human compare PDF/video/studio; don't import full score into app
OPEN-07	How to provide lawful high-quality reference and original A/B audio with minimal architecture	Product/copyright/engineering design choice, not yet committed
OPEN-08	Is a saved Principle Card useful to the learner after a first session?	Physical pilot observation before building persistence
OPEN-09	Should first release include song overview of all sections or only entry to one curated passage with section context?	Product owner prioritization
OPEN-10	Independent physical guitar test, on mobile and desktop, not yet conducted	Must precede any learner-facing merge

ID	Open issue	Impact / resolution owner
OPEN-11	Require actual checks of accessibility/tempo accuracy before quantitative performance targets	Engineering gate for relevant future slice

12.1 Proposed approval decisions (none approved yet)

1. **D-01:** Retain song-led study with optional transfer to general musicianship as the north star, rather than restoring independent exercise menus.
2. **D-02:** Accept the 11 macro-section provisional structure **for research**, without freezing its event count or timestamps as truth.
3. **D-03:** Approve or change the **single-fragment Chromatic Launchpad** as first vertical slice, versus Argon's larger 3-investigation bundle.
4. **D-04:** Confirm one main workbench with conditional views, and a separate song overview.
5. **D-05:** Require sourced vs. inferred musical claims to remain separate, reusing the existing research contract where suitable.
6. **D-06:** Choose minimal reference strategy for the pilot: external studio/tutorial links plus lawful, original educational A/B sound if needed; **no automatic 3-stem player commitment**.
7. **D-07:** Use physical-guitar self-report and qualitative learning evidence first; defer performance grading, microphone mirroring and mandatory recording.
8. **D-08:** Defer saved Principle Cards, universal investigation templates, imported TAB and Phrase v1 modifications until actual need is shown.
9. **D-09:** Resolve PR #84 and preview gate independently from Song Study design approval.

13. Exit criteria for this FRD draft

To move from v0.1 to v0.2: (a) review the 11-section map against the official studio recording; (b) fix flagged harmony/time claims; (c) agree the chosen pilot experience and source/audio strategy; (d) inspect the pilot against current engine APIs and research provenance contract; (e) record first physical-guitar test plan; (f) obtain owner confirmation of the revised requirements and separately resolve cleanup PR #84.

To authorize any Song Study implementation: a new, bounded GitHub issue/PR must reference an approved FRD revision and exact accepted master base. This draft is expressly **not** that authorization.

Appendix A. Argon-to-FRD correction ledger

Correction family	Applied in v0.1	Open verification
COR-MAP-01–03	11 macro-section / 31 atomic event *draft*; bridge separated from chordal chorus; solo pickup not presented as settled studio fact	Independently check timestamps, bridge material and solo entry against recording
COR-RHY-01–04	Explicit eighth-note subdivision and beat length, 1+1+1+1+1+2+2 pattern, pickup distinguished from drum-bar downbeat	Reconstruct 9+9+9+5 phase against recording/video before precise learner UI
COR-THY-01–06	Ascending chromatic approach terminology, pitch collection/scale distinction, power-chord precision and conditional harmonic labels	Source-backed bridge / outro analysis, chromatic embellishments, studio vs live variants
COR-SRC-01–03	Distinct competitor features, source attribution and existing src/research/ preservation verified from PR #84	Confirm source-specific citations and exact use rights before production publication

Appendix B. Research vs. product authority

This document is a human-facing specification derived from two supplied Argon reports, the user-confirmed Marty tutorial and uploaded solo PDF, and read-only GitHub checks at PR #84's proposed baseline. Argon's “verified” designations are **claims of the research source**, not a substitute for our own final audio, notation and learner validation. No source supports automatic assessment of the owner's real guitar performance, a universal seven-template architecture, or unrestricted reproduction of Led Zeppelin music.